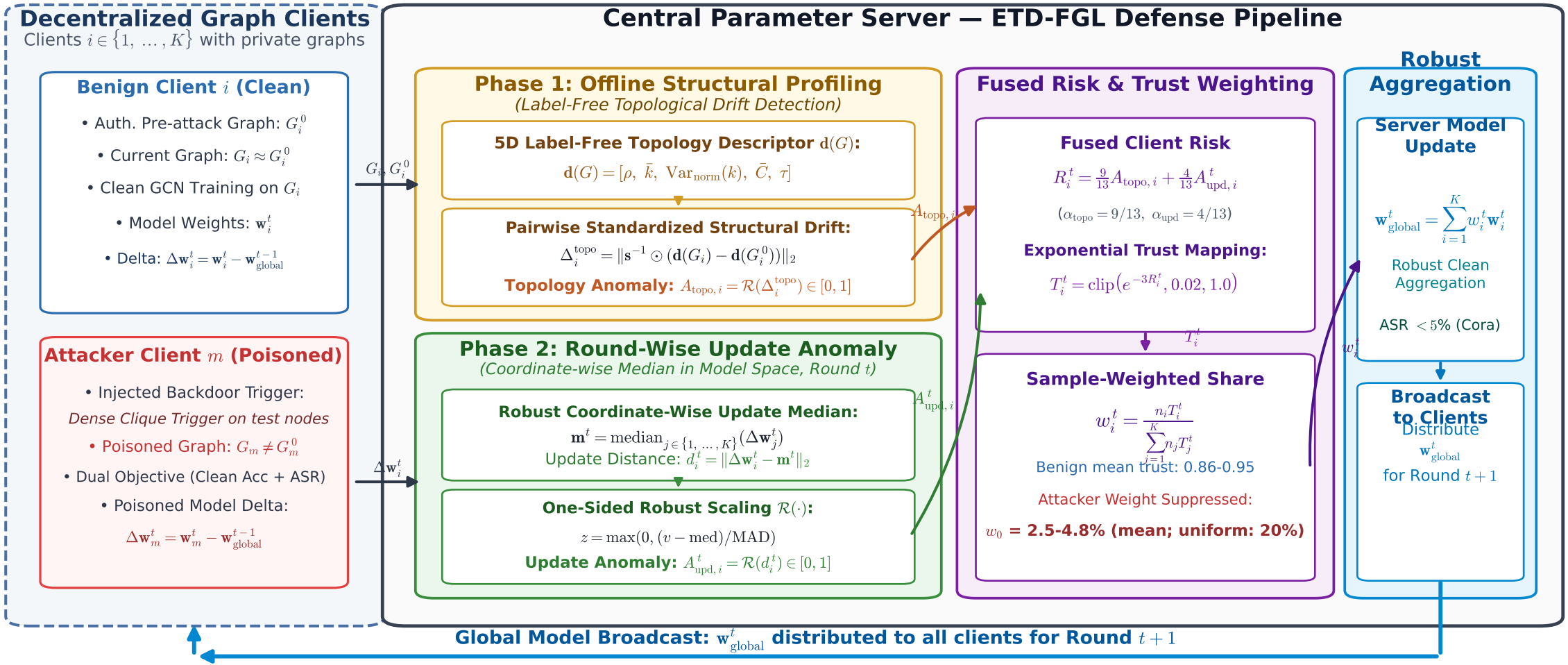

Decentralized Graph Clients
Clients $i \in \{1, \ldots, K\}$ with private graphs
Benign Client i (Clean)
• Auth. Pre-attack Graph: G_i^0
• Current Graph: $G_i \approx G_i^0$
• Clean GCN Training on G_i
• Model Weights: $\mathbf{w}_i^t$
• Delta: $\Delta \mathbf{w}_i^t = \mathbf{w}_i^t - \mathbf{w}_{\text{global}}^{t-1}$
Attacker Client m (Poisoned)
• Injected Backdoor Trigger:
Dense Clique Trigger on test nodes
• Poisoned Graph: $G_m \neq G_m^0$
• Dual Objective (Clean Acc + ASR)
• Poisoned Model Delta:
$\Delta \mathbf{w}_m^t = \mathbf{w}_m^t - \mathbf{w}_{\text{global}}^{t-1}$
Central Parameter Server — ETD-FGL Defense Pipeline
G_i, G_i^0
$\Delta \mathbf{w}_i^t$
Phase 1: Offline Structural Profiling
(Label-Free Topological Drift Detection)
5D Label-Free Topology Descriptor $\mathbf{d}(G)$:
$\mathbf{d}(G) = [\rho,\ \bar{k},\ \mathrm{Var}_{\text{norm}}(k),\ \bar{C},\ \tau]$
Pairwise Standardized Structural Drift:
$\Delta_i^{\text{topo}} = \|\mathbf{s}^{-1} \odot (\mathbf{d}(G_i) - \mathbf{d}(G_i^0))\|_2$
Topology Anomaly: $A_{\text{topo},i} = \mathcal{R}(\Delta_i^{\text{topo}}) \in [0,1]$
$A_{\text{topo},i}$
Phase 2: Round-Wise Update Anomaly
(Coordinate-wise Median in Model Space, Round t)
Robust Coordinate-Wise Update Median:
$\mathbf{m}^t = \text{median}_{j \in \{1,\ldots,K\}}(\Delta \mathbf{w}_j^t)$
Update Distance: $d_i^t = \|\Delta \mathbf{w}_i^t - \mathbf{m}^t\|_2$
One-Sided Robust Scaling $\mathcal{R}(\cdot)$:
$z = \max(0, (v - \text{med})/\text{MAD})$
Update Anomaly: $A_{\text{upd},i}^t = \mathcal{R}(d_i^t) \in [0,1]$
$A_{\text{upd},i}^t$
Fused Risk & Trust Weighting
Fused Client Risk
$R_i^t = \frac{9}{13} A_{\text{topo},i} + \frac{4}{13} A_{\text{upd},i}^t$
$(\alpha_{\text{topo}} = 9/13,\ \alpha_{\text{upd}} = 4/13)$
Exponential Trust Mapping:
$T_i^t = \text{clip}(e^{-3R_i^t}, 0.02, 1.0)$
T_i^t
Sample-Weighted Share
$w_i^t = \frac{n_i T_i^t}{\sum_{j=1}^{K} n_j T_j^t}$
Benign mean trust: 0.86-0.95
Attacker Weight Suppressed:
w_0 = 2.5-4.8% (mean; uniform: 20%)
Robust Aggregation
Server Model Update
$\mathbf{w}_{\text{global}}^t = \sum_{i=1}^{K} w_i^t \mathbf{w}_i^t$
Robust Clean Aggregation
ASR $<5\%$ (Cora)
w_i^t
Broadcast to Clients
Distribute $\mathbf{w}_{\text{global}}^t$
for Round $t+1$
Global Model Broadcast: $\mathbf{w}_{\text{global}}^t$ distributed to all clients for Round $t+1$